⚡ SSC CGL Speed Book — Tricks & Shortcuts

21 high-value tricks across 8 topics. Each card gives the **shortcut · mental calculation · when to use · when NOT to use · traps · PYQ · where it fails**.

How to revise: read the shortcut, cover the card, redo the PYQ in your head, and re-read *where it fails* — that line is what saves marks.

🧭 Contents

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Percentage ↑ top

PCT-01 Fraction ⇄ Percentage memorised table ★

✅ SSC Shortcut	Memorise 1/n as a %: 1/2=50, 1/3=33.33, 1/4=25, 1/5=20, 1/6=16.66, 1/7=14.28, 1/8=12.5, 1/9=11.11, 1/11=9.09, 1/12=8.33. Then x/n is just a multiple.
🧮 Mental Calculation	To find 37.5% of 240: 37.5% = 3/8, so 240÷8×3 = 30×3 = 90. Converting the % to a small fraction makes it a one-step division.

🔎 **When to Use**	Whenever the percentage equals a clean fraction (12.5, 16.66, 37.5, 62.5, 87.5, 33.33, 66.66) — extremely common in profit-loss, DI, and data questions.
🚫 **When NOT to Use**	When the percentage is an 'ugly' number (e.g. 23%, 47%) that is not a neat fraction — then plain multiplication is faster than forcing a fraction.
⚠️ **Common Traps**	Mixing up 1/6 = 16.66% with 1/16 = 6.25%. And 0.66 is 66.66%, not 66%.
📝 **PYQ Example**	Q: 87.5% of 64 = ? → 87.5% = 7/8 → 64÷8×7 = 8×7 = 56. Answer: 56.
❌ **Where the Shortcut Fails**	Fails for repeating decimals when the question needs exact rupees (e.g. 1/3 of ₹100 = ₹33.33…) — round only at the final step, or the running error compounds.

PCT-02 Successive % change = a + b + ab/100 ★

✅ **SSC Shortcut**	Two successive changes of a% then b% give a single net change of (a + b + ab/100)%. Use −b for a decrease.
🧮 **Mental Calculation**	20% up then 10% down: 20 − 10 − 200/100 = 10 − 2 = +8%. Add the two, then add their product over 100.
🔎 **When to Use**	Price first raised then lowered, population grows two years, length×breadth area change, two successive discounts.
🚫 **When NOT to Use**	When more than 2 stages and you need an exact figure — chaining multipliers (×1.2×0.9) is cleaner than repeating the formula.
⚠️ **Common Traps**	Forgetting the ab/100 term and just doing a+b. And the sign: a decrease must enter as negative in BOTH the sum and the product.
📝 **PYQ Example**	Q: A salary rises 10% then falls 10%. Net? → 10 − 10 − 100/100 = −1% (a 1% loss, never zero).
❌ **Where the Shortcut Fails**	Doesn't directly give the rupee value — only the %. If the base differs between the two stages (not the same quantity), the formula is invalid.

PCT-03 Product constancy (if one ↑ x%, other ↓ to keep product same) ★

✅ SSC Shortcut	If A×B is fixed and A rises by x%, B must fall by x/(100+x)×100 %. If A falls by x%, B rises by x/(100−x)×100 %.
🎛️ Mental Calculation	Price up 25% (=1/4), so to keep expenditure same, consumption down 1/(4+1) = 1/5 = 20%. Convert x% to a fraction, add/subtract 1 in the denominator.
🔎 When to Use	Price–consumption–expenditure, speed–time (fixed distance), length–breadth (fixed area) questions.
🚫 When NOT to Use	When the product itself changes (e.g. expenditure also increases by some %) — then it's a normal successive-change problem, not constancy.
⚠️ Common Traps	Using x/(100) instead of x/(100±x). The reduction % is NOT equal to the increase %.
📝 PYQ Example	Q: Petrol price rises 20%. By what % must use drop to keep the bill same? → 20/(120)×100 = 16.66%.
❌ Where the Shortcut Fails	Fails if the two quantities are not in pure inverse proportion (e.g. a fixed charge is added) — the constancy assumption breaks.

Profit & Loss ↑ top

PL-01 Single-multiplier chaining ★

✅ SSC Shortcut	Treat every deal as a multiplier on CP: +p% → ×(1+p/100), −l% → ×(1−l/100). Chain all deals, then solve once.
🎛️ Mental Calculation	A→B +20%, B→C +25%: 1.2×1.25 = 1.5. So C pays 1.5× A's cost. Multiply factors, don't add percentages.
🔎 When to Use	An article passes through a chain of buy–sell deals; or CP↔SP at a given %.
🚫 When NOT to Use	Single straightforward CP→SP — just use SP = CP(100±p)/100 directly; chaining is overkill.

⚠️ Common Traps	Adding the percentages (20+25 = 45%) instead of multiplying the factors (×1.5 = 50%).
📝 PYQ Example	Q: D pays ₹1350 after A→B +20%, B→C +25%, C→D −10%. A's cost? → 1.2×1.25×0.9 = 1.35 → 1350/1.35 = ₹1000.
❌ Where the Shortcut Fails	Fails when a deal is on a DIFFERENT base (e.g. profit on SP, not CP) — convert that deal's base first or the multiplier is wrong.

PL-02 Two articles at same SP, ±x% → always loss $(x/10)^2$% ★

✅ SSC Shortcut	Sell two items at the SAME price, one at +x% and one at −x%: net is ALWAYS a loss of (x/10)% squared = $x^2/100$ %.
🧮 Mental Calculation	x = 10 → loss = 1%. x = 20 → loss = 4%. x = 25 → 6.25%. Just halve-ish: (x/10) then square.
🔎 When to Use	'Two items sold at the same price, one gains x%, other loses x%' — asks net profit/loss%.
🚫 When NOT to Use	If the two SELLING PRICES are different, or the gain% ≠ loss% — then compute CPs separately; the formula is invalid.
⚠️ Common Traps	Assuming gain and loss cancel to 0%. They never do at equal SP — it's always a loss.
📝 PYQ Example	Q: Two scooters sold at ₹1.4L each, +25% and −25%. Net? → loss $(25/10)^2$ = 6.25%.
❌ Where the Shortcut Fails	Fails the moment the percentages differ (e.g. +20% and −25%) — there is no clean square formula then.

PL-03 Dishonest dealer false-weight gain %

✅ SSC Shortcut	Sells at CP but uses a false weight: Gain% = error/(true − error) × 100, where 'error' = shortfall in grams.
🧮 Mental Calculation	Uses 900 g for 1 kg → error 100, true 1000: 100/900×100 = 11.11% gain. Shortfall over the weight actually given.
🔎 When to Use	Shopkeeper claims to sell at cost price but gives less than the stated weight.

🚫 When NOT to Use	When the dealer ALSO marks up the price — then combine the markup multiplier with the weight gain (two effects).
⚠️ Common Traps	Dividing by 1000 (true weight) instead of 900 (given weight). Denominator is what the customer actually receives.
📝 PYQ Example	Q: Uses a 950 g weight for a kg, sells at CP. Gain%? → 50/950×100 = 5.26%.
❌ Where the Shortcut Fails	Fails if the goods are also sold at a profit/discount — the single formula only covers the weight cheat alone.

Average & Mixtures ↑ top

AVG-01 Average speed for equal distance (harmonic mean) ★

✅ SSC Shortcut	Equal distance at speeds a and b → average speed = 2ab/(a+b), NOT (a+b)/2.
🧮 Mental Calculation	60 and 40: 2×60×40/100 = 4800/100 = 48 km/h. Product×2 over the sum.
🔎 When to Use	Same distance covered at two different speeds (go at a, return at b).
🚫 When NOT to Use	When the TIME is equal (not distance) — then it IS the simple mean (a+b)/2.
⚠️ Common Traps	Using the simple average (a+b)/2 = 50 — wrong; equal distance always gives a number below the simple mean.
📝 PYQ Example	Q: 120 km at 60, back at 40. Avg speed? → 2·60·40/100 = 48 km/h.
❌ Where the Shortcut Fails	Fails for three speeds or unequal distances — use total distance ÷ total time instead.

AVG-02 Alligation cross method ★

✅ SSC Shortcut	Mix cheap (c) and dear (d) to get mean (m): ratio cheap:dear = (d − m) : (m − c). Cross the differences.

🧮 Mental Calculation	Tea ₹12 & ₹18 to make ₹16: (18−16):(16−12) = 2:4 = 1:2. The closer price needs the larger share.
🔎 When to Use	Two ingredients/groups combined to a known average; finding the mix ratio.
🚫 When NOT to Use	Three or more ingredients with no pairing — split into pairs or solve by equations.
⚠️ Common Traps	Putting the ratio on the wrong side. The quantity is INVERSELY proportional to the distance from the mean.
📝 PYQ Example	Q: In what ratio mix ₹20 and ₹30 rice to sell at ₹24 (cost)? → (30−24):(24−20) = 6:4 = 3:2.
❌ Where the Shortcut Fails	Fails when the 'mean' is outside [c, d] — impossible to mix to it; check the mean lies between the two.

AVG-03 Repeated replacement / dilution ★

✅ SSC Shortcut	Remove x units and refill with water, n times, from a vessel of volume V: pure liquid left = Original × $(1 - x/V)^n$.
🧮 Mental Calculation	20 L milk, remove 4 L & refill, twice: $20\times(1-4/20)^2 = 20\times(0.8)^2 = 20\times0.64 = 12.8$ L milk left.
🔎 When to Use	Same amount drawn off and replaced by water repeatedly; find remaining pure quantity/ratio.
🚫 When NOT to Use	When DIFFERENT amounts are removed each round — apply the multiplier stage by stage, not the single power.
⚠️ Common Traps	Using x/Original instead of x/V (vessel volume), and forgetting the power n.
📝 PYQ Example	Q: 40 L, 8 L replaced thrice. Milk left? → $40\times(1-8/40)^3 = 40\times(0.8)^3 = 40\times0.512 = 20.48$ L.
❌ Where the Shortcut Fails	Fails if water is added WITHOUT first removing mixture (volume grows) — that's a different ratio problem.

Simple & Compound Interest ↑ top

SICI-01 CI − SI difference (2 & 3 years) ★

✅ SSC Shortcut	For 2 years: CI − SI = $P(R/100)^2$. For 3 years: CI − SI = $P(R/100)^2 \cdot (3 + R/100)$.
🧮 Mental Calculation	P=8000, R=5%, 2 yr: $8000 \times (1/20)^2$ = 8000/400 = ₹20. Square the rate-fraction, times P.
🔎 When to Use	Question gives the CI−SI gap (or asks for it) over 2 or 3 years at the same rate.
🚫 When NOT to Use	For 1 year (CI = SI, difference 0), or when compounding is half-yearly — adjust R and n first.
⚠️ Common Traps	Using R/100 without squaring; or applying the 2-year formula to a 3-year gap.
📝 PYQ Example	Q: Difference between CI and SI on ₹10000 at 10% for 2 yrs? → $10000 \times (0.1)^2$ = ₹100.
❌ Where the Shortcut Fails	Fails for non-annual compounding (half-yearly/quarterly) unless you convert rate and time accordingly.

SICI-02 Doubling time — Rule of 72 (CI)

✅ SSC Shortcut	Money doubles under compound interest in ≈ 72/R years. Under simple interest it doubles in exactly 100/R years.
🧮 Mental Calculation	At 8% CI, doubles in 72/8 = 9 years. At 8% SI, doubles in 100/8 = 12.5 years.
🔎 When to Use	Quick estimate of doubling/tripling time, or back-solving the rate from a doubling period.
🚫 When NOT to Use	When an EXACT compound answer is needed — Rule of 72 is an approximation; use $(1+R/100)^n$ for precision.
⚠️ Common Traps	Applying 72/R to simple interest (it's for compound). For SI, doubling is 100/R, tripling 200/R.
📝 PYQ Example	Q: A sum doubles in 4 yrs at SI. Rate? → 100/4 = 25% per annum.

❌ Where the Shortcut Fails	Inaccurate at high rates (e.g. 24%+) and for non-doubling targets; it's an estimate, not exact for CI.

Time & Work ↑ top

TW-01 LCM method (assume total work = LCM of days) ★

✅ SSC Shortcut	Let total work = LCM of the given days. Each worker's daily output = total ÷ his days. Add rates, divide back.
🧮 Mental Calculation	A=10 d, B=15 d → total = LCM 30. A=3/day, B=2/day → together 5/day → 30/5 = 6 days.
🔎 When to Use	Almost every time-and-work problem with combined/alternate workers, leaving, or joining.
🚫 When NOT to Use	Pure single-worker proportion ($M_1D_1=M_2D_2$) is faster for simple men-days scaling.
⚠️ Common Traps	Forgetting to add efficiencies before dividing; or using days instead of rates when combining.
📝 PYQ Example	Q: A 12 d, B 18 d, together? → LCM 36; rates 3+2=5; 36/5 = 7.2 days.
❌ Where the Shortcut Fails	Gets messy when efficiencies are given as ratios with no days, or work is done in unequal shifts — model rates directly.

TW-02 Efficiency ratio ⇒ inverse time ratio

✅ SSC Shortcut	If A is x times as efficient as B, A takes 1/x the time. Efficiency and time are inversely proportional.
🧮 Mental Calculation	A is 3× as efficient as B and they differ by 20 days: time ratio 1:3, difference 2 parts = 20 → 1 part 10 → A=10, B=30 days.
🔎 When to Use	'A is twice as good as B', or 'A does in 10 days what B does in 15' type comparisons.
🚫 When NOT to Use	When absolute work rates are already given — just use the LCM method.

⚠️ Common Traps	Flipping the ratio: more efficient = LESS time, so the efficient worker gets the SMALLER time share.
📝 PYQ Example	Q: A is thrice as efficient as B; B takes 60 days. A? → 60/3 = 20 days.
❌ Where the Shortcut Fails	Fails if efficiency changes mid-task (fatigue, machine slowdown) — constant-rate assumption breaks.

Time, Speed & Distance ↑ top

TSD-01 km/h ⇄ m/s with 5/18 ★

✅ SSC Shortcut	km/h → m/s: multiply by 5/18. m/s → km/h: multiply by 18/5.
🧮 Mental Calculation	72 km/h → 72×5/18 = 4×5 = 20 m/s. (72/18 = 4, then ×5.)
🔎 When to Use	Any train/relative-speed problem mixing km/h with seconds and metres.
🚫 When NOT to Use	When all units are already consistent (all km & h, or all m & s) — no conversion needed.
⚠️ Common Traps	Using 18/5 the wrong direction. km/h is the bigger-feeling unit; to get m/s the number gets SMALLER (×5/18).
📝 PYQ Example	Q: A train at 90 km/h crosses a pole in 8 s. Length? → 90×5/18 = 25 m/s; 25×8 = 200 m.
❌ Where the Shortcut Fails	No failure case — just keep the direction straight; the trap is purely the multiplier orientation.

TSD-02 Train crossing: pole vs platform ★

✅ SSC Shortcut	Crossing a pole (point): time = train length ÷ speed. Crossing a platform/bridge: time = (train + platform length) ÷ speed.
🧮 Mental Calculation	A pole adds nothing; a platform adds its own length to the distance. Decide first whether the object has length.
🔎 When to Use	Train passing a pole/man (point object) vs a platform/tunnel/another train (has length).

🚫 When NOT to Use	Two trains crossing each other — use relative speed (sum if opposite, difference if same direction) plus both lengths.
⚠️ Common Traps	Adding the platform length when crossing a POLE, or forgetting to add the train's own length for a platform.
📝 PYQ Example	Q: 150 m train at 20 m/s crosses a 250 m platform. Time? → (150+250)/20 = 20 s.
❌ Where the Shortcut Fails	Fails for two moving trains — that needs relative speed, not this single-object formula.

TSD-03 Boats & streams half-sum / half-difference

✅ SSC Shortcut	Downstream = b+s, Upstream = b−s. So boat speed b = (down+up)/2 and stream speed s = (down−up)/2.
🧮 Mental Calculation	Down 12, up 8: boat = (12+8)/2 = 10, stream = (12−8)/2 = 2 km/h. Average for boat, half-gap for stream.
🔎 When to Use	Given downstream and upstream speeds (or times over a fixed distance), find boat or stream speed.
🚫 When NOT to Use	When only one of down/up is given without the other — you need both, or a distance-time pair.
⚠️ Common Traps	Swapping which is bigger: downstream is ALWAYS faster (b+s). Stream speed can't exceed boat speed.
📝 PYQ Example	Q: Down 15 km/h, up 9 km/h. Stream speed? → (15−9)/2 = 3 km/h.
❌ Where the Shortcut Fails	Fails if the stream speed varies along the route or the distances up/down differ — model with time = d/speed.

Algebra ↑ top

ALG-01 x + 1/x identities ★

✅ SSC Shortcut	If $x + 1/x = k$: then $x^2 + 1/x^2 = k^2 - 2$, and $x^3 + 1/x^3 = k^3 - 3k$. If $x - 1/x = k$: $x^2 + 1/x^2 = k^2 + 2$.

🧮 **Mental Calculation**	$x+1/x = 3 \rightarrow x^2+1/x^2 = 9-2 = 7 \rightarrow x^3+1/x^3 = 27-9 = 18$. Square minus 2, then cube minus 3k.
🔎 **When to Use**	Given a sum/difference of a number and its reciprocal; asked higher powers.
🚫 **When NOT to Use**	When x itself is asked (solve the quadratic) rather than a symmetric power expression.
⚠️ **Common Traps**	Using k^2+2 for the $(x+1/x)$ case — it's k^2-2 for a sum, k^2+2 for a difference.
📝 **PYQ Example**	Q: If $x+1/x = 2$, find $x^3+1/x^3 \rightarrow 2^3 - 3\cdot2 = 8 - 6 = 2$.
❌ **Where the Shortcut Fails**	Doesn't help for x + 1/y (different variables) or non-reciprocal pairs — identities assume the exact reciprocal.

ALG-02 $a+b+c = 0 \Rightarrow a^3+b^3+c^3 = 3abc$

✅ **SSC Shortcut**	Whenever a + b + c = 0, the sum of cubes collapses to 3abc.
🧮 **Mental Calculation**	Spot that the three terms add to zero, then just multiply them and triple it — skip all expansion.
🔎 **When to Use**	Three quantities visibly summing to zero (e.g. $(x-y)+(y-z)+(z-x) = 0$).
🚫 **When NOT to Use**	When $a+b+c \neq 0$ — must use the full identity $a^3+b^3+c^3-3abc = (a+b+c)(a^2+b^2+c^2-ab-bc-ca)$.
⚠️ **Common Traps**	Applying 3abc without first CHECKING that the sum is actually zero.
📝 **PYQ Example**	Q: If a=7, b=−3, c=−4 (sum 0), $a^3+b^3+c^3 = 3abc = 3\cdot7\cdot(-3)\cdot(-4) = 252$.
❌ **Where the Shortcut Fails**	Completely invalid the instant the three don't sum to zero — always verify the condition.

ALG-03 $x=(a+b)/(a-b)$, $y=(a-b)/(a+b)$ reciprocal pair ★

✅ **SSC Shortcut**	If $x=(a+b)/(a-b)$ and $y=(a-b)/(a+b)$, then instantly: $xy=1$ (they're reciprocals) and $x+y = 2(a^2+b^2)/(a^2-b^2)$. Then $x^2+y^2 = (x+y)^2-2xy = (x+y)^2-2$.

🧮 **Mental Calculation**	Write $xy=1$ first. For $x+y$, top = $2(a^2+b^2)$, bottom = a^2-b^2. Then square the sum and subtract 2 — no fraction expansion at all.
🔎 **When to Use**	Question gives x and y as $(a+b)/(a-b)$ and its flip, and asks $x+y$, x^2+y^2, x^3+y^3, or $1/x+1/y$.
🚫 **When NOT to Use**	When x and y are NOT exact reciprocals (e.g. y has a different numerator) — then $xy \neq 1$ and you must compute both separately.
⚠️ **Common Traps**	Forgetting that $xy=1$ makes $1/x+1/y = x+y$, and $x^2+y^2 = (x+y)^2-2$ (not $-2xy$ left as a variable). Also a^2-b^2 in the denominator can be negative if $b>a$ — keep the sign.
📝 **PYQ Example**	Q: $a=5, b=3 \Rightarrow x=4, y=1/4$. $x+y=2(25+9)/(25-9)=68/16=4.25$; $x^2+y^2=4.25^2-2=18.0625-2=16.0625$. ✓
❌ **Where the Shortcut Fails**	Fails if $a=b$ (denominator $a-b=0$, undefined) or if the pair isn't truly reciprocal — verify $xy=1$ before using $x^2+y^2=(x+y)^2-2$.

Geometry & Mensuration ↑ top

GEO-01 Pythagorean triples (skip the root) ★

✅ **SSC Shortcut**	Recognise triples instead of computing roots: 3-4-5, 5-12-13, 8-15-17, 7-24-25, 9-40-41, 20-21-29 (and their multiples).
🧮 **Mental Calculation**	Sides 9 and 12 → it's 3-4-5 ×3 → hypotenuse 15. Spot the family, scale up.
🔎 **When to Use**	Right triangles, diagonals, ladder/wall, and any $\sqrt{(a^2+b^2)}$ where a,b fit a known triple.
🚫 **When NOT to Use**	When the sides don't form a triple — then actually compute $\sqrt{(a^2+b^2)}$.
⚠️ **Common Traps**	Assuming every right triangle is 3-4-5; check the ratio first. 6-8-10 is a triple, 6-8-9 is not.
📝 **PYQ Example**	Q: Ladder foot 5 m from wall, top at 12 m. Length? → 5-12-13 → 13 m.

❌ **Where the Shortcut Fails**	Fails for non-triple sides (e.g. 2 and 3) — no clean integer hypotenuse, must use the root.

GEO-02 Painted cube — 3/2/1/0 face counts

✅ **SSC Shortcut**	Cube painted then cut into n^3: 3 faces = 8 (corners), 2 faces = $12(n-2)$, 1 face = $6(n-2)^2$, 0 faces = $(n-2)^3$.
🎛️ **Mental Calculation**	$n=4$: corners 8, edges $12 \cdot 2=24$, faces $6 \cdot 4=24$, inner $2^3=8$. They sum to $64 = 4^3$ (always check).
🔎 **When to Use**	A painted cube cut into equal smaller cubes; asks how many have k painted faces.
🚫 **When NOT to Use**	When only some faces are painted, or two faces left blank — recompute; the standard counts assume all 6 painted.
⚠️ **Common Traps**	Using n instead of $(n-2)$ for edges/faces/inner. Corners are always 8 regardless of n.
📝 **PYQ Example**	Q: Cube cut into 27 ($n=3$), how many have exactly 2 faces painted? → $12(3-2) = 12$.
❌ **Where the Shortcut Fails**	Fails when faces are partially painted or the cuts are unequal — the symmetric formula no longer holds.

Goal: 200–300 tricks by exam time. Add new ones to `speedbook.json` (or `new_tricks.json`) and run `py speedbook.py`.